Computer Vision

Kenneth.Dawson-Howe@scss.tcd.ie

School of Computer Science and Statistics,

Trinity College, University of Dublin. Ireland .

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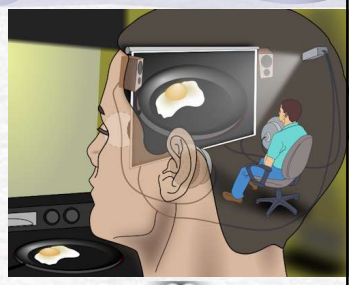
Module contents

Week	Beginning	Folder	Topic	Format	Mini Tests (10%)	Live Classes	Practical Component
1	14/09/2026	Introduction	What is Computer Vision	Live		Lecture Tuesday @2pm	Install OpenCV & TIPS.
		Introduction	Module Organisation	Live		Lecture Wednesday @1pm	Experiment with TIPS
2	21/09/2026	Images	Camera Model & Digitisation	Pre-recorded			
		Images	Colour Image models	Pre-recorded			
		Images	Noise & Smoothing	Pre-recorded	Images before Tuesday @12noon	Q&A Session Tuesday @2pm	
		Binary	Thresholding	Pre-recorded		Tutorial Wednesday @1pm	
		Binary	Cleaning Binary Images	Pre-recorded	Binary before Tuesday @12noon		
3	28/09/2026	Histograms	1D & 3D histograms	Pre-recorded			
		Histograms	Equalisation, Comparison & Back Projection	Pre-recorded	Histograms before Tuesday @12noon	Q&A Session Tuesday @2pm	
		Regions	Connectivity & Connected Component Analysis	Pre-recorded		Tutorial Wednesday @1pm	
		Regions	Features	Pre-recorded			
		Regions	kMeans clustering	Pre-recorded	Regions before Tuesday @12noon		
4	05/10/2026	Edges	Edge detection	Pre-recorded			Assignment (30%)
		Edges	Contour segmentation	Pre-recorded		Q&A Session Tuesday @2pm	- Released on 7-Oct @2.00pm
		Edges	Hough transform	Pre-recorded		Tutorial Wednesday @1pm	
		Edges	Least squared error & RANSAC	Pre-recorded	Edges before Tuesday @12noon		
5	12/10/2026	Geometric	Transformations	Pre-recorded			
		Geometric	Interpolation	Pre-recorded	Geometric before Tuesday @12noon	Q&A Session Tuesday @2pm	
		Learning & Evaluation	Learning & Ground Truth	Pre-recorded		Tutorial Wednesday @1pm	
		Learning & Evaluation	Performance	Pre-recorded	Learning & Evaluation before Tuesday @12noon		
6	19/10/2026	Regions II	Watershed segmentation	Pre-recorded			
		Regions II	Mean shift	Pre-recorded	Regions II before Tuesday @12noon	Q&A Session Tuesday @2pm	
		Video	Introduction	Pre-recorded		Tutorial Wednesday @1pm	
		Video	Background Models	Pre-recorded	Video before Tuesday @12noon		
8	02/11/2026	Features	Corner detection - Moravec	Pre-recorded			
		Features	Corner detection - Harris	Pre-recorded		Q&A Session Tuesday @2pm	
		Features	SIFT Keypoints	Pre-recorded		Tutorial Wednesday @1pm	
		Features	SIFT Matching & Recognition	Pre-recorded	Features before Tuesday @12noon		
9	09/11/2026	Video II	Tracking - Mean Shift	Pre-recorded			
		Video II	Tracking - Mean Shift	Pre-recorded		Q&A Session Tuesday @2pm	
		Video II	Tracking - Optical Flow	Pre-recorded	Video II before Tuesday @12noon	Tutorial Wednesday @1pm	Submit Assignment
							- before 11-Nov @11.59pm
10	16/11/2026	Recognition	Template Matching	Pre-recorded			
		Recognition	Chamfer Matching	Pre-recorded		Q&A Session Tuesday @2pm	
		Recognition	Haar	Pre-recorded		Tutorial Wednesday @1pm	
		Recognition	Haar	Pre-recorded			
		Recognition	PCA	Pre-recorded	Recognition before Tuesday @12noon		
11	23/11/2026	Deep Learning	Linear Classifiers	Pre-recorded			
		Deep Learning	Neural Networks	Pre-recorded		Q&A Session Tuesday @2pm	
		Deep Learning	Convolution Networks	Pre-recorded	Deep Learning before Tuesday @12noon	Tutorial Wednesday @1pm	
		Deep Learning	ResNet, Transfer, RNN, R-CNN	Pre-recorded			
12	30/11/2026	Conclusions	How to address vision problems	Live		Lecture Tuesday @2pm	
		Conclusions	Sample exam paper tutorial	Live		Tutorial Wednesday @1pm	
							Exam (60%)

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Introduction – Human Vision

- When it was discovered that an image is projected inside the eye…
  - Scientists thought vision was solved.
  - But there is no homunculus in our brain.
- So how do we understand the images from the eyes?
  - 20% of the brain does "visual only" processing.
  - 40%+ of the brain does "vision+" processing.
    - Motor, attention, navigation, …
- What does the brain do?
  - Visual cortex – basic image (pre)processing
  - Dedicated areas – face recognition, motion analysis, …
  - We know very little really.



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Courtesy of Pbroks13 and Jennifer Garcia, see Wikipedia article on Cartesian theatre for more information

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Introduction – Emulating human vision

- Machines can easily solve many tasks that we find difficult.
  - Particularly if we can describe them mathematically or in terms of an algorithm
- The solutions of many tasks that we find easy seem very difficult to describe formally.
  - Object recognition.
    - Rigid objects (variety)
    - Deformable objects
  - World modelling.



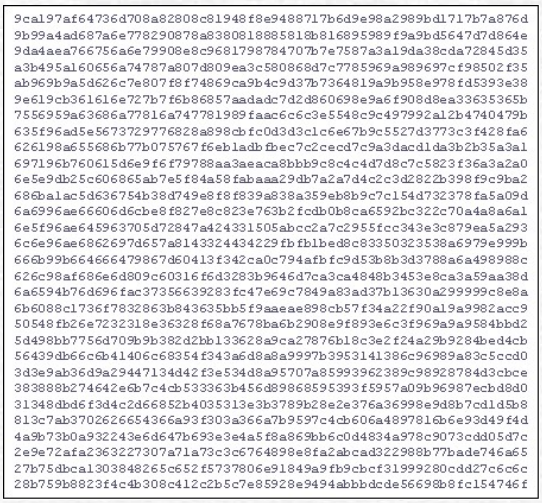
- Is machine vision just implementing human vision on a machine?
  - We don't understand human vision properly.
  - Machine vision is not restricted to two images from the visible spectrum.
    - UV, IR, Multiple cameras, Laser range finders, …
  - Machine learning can "tackle tasks that are too difficult to solve with fixed programs written and designed by human beings." From *Deep Learning* by Goodfellow

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Computer Vision is hard

"What we experience, apparently directly, is actually very different from what is recorded by our sense organs."

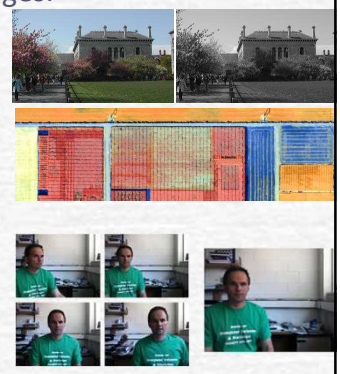
[Perception: From Sense to Object by J. Wilding]



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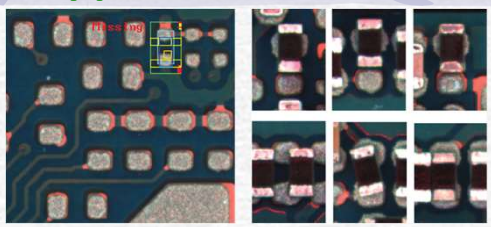
Introduction

- Computer Vision is about understanding images. These images can be
  - Greyscale or Colour or Multi-spectral
  - Snapshots or video sequences
  - Taken with a static or moving camera
  - Taken of a stationary or dynamic scene
  - Taken with a calibrated or un-calibrated camera
- What Computer Vision aims to do is to extract some useful information from these images for…
  - Inspection purposes
  - Analysis purposes
  - Control purposes



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Applications – Industrial Inspection



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Applications – Surveillance / Forensics

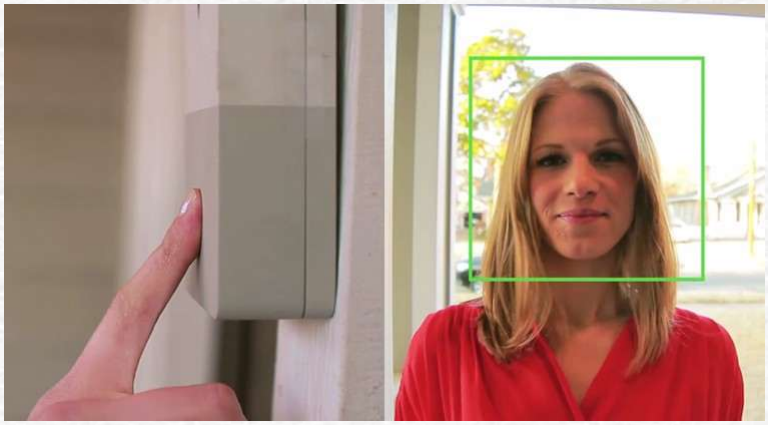
- In London there are almost 1,000,000 cameras (SLT, 2025)
  - Average person caught on camera 100-500 times per day.
  - 13% reduction in crime?
- However, only 1 crime solved per 1,000 cameras (BBC, 2009)
- In July 2005 there was a major terrorist attack in London…
- Afterwards
  - 17,000 hours of video was studied
  - Over 1 year



© Image courtesy of Feelart / FreeDigitalPhotos.net

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Applications – Biometrics

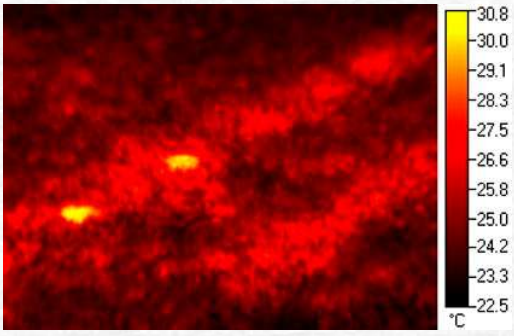


No need for a key. Just your face. Won't be fooled by photos.

Chui Smart Doorbell from http://iothought.com/chui-smart-doorbell/

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Applications – Landmine detection

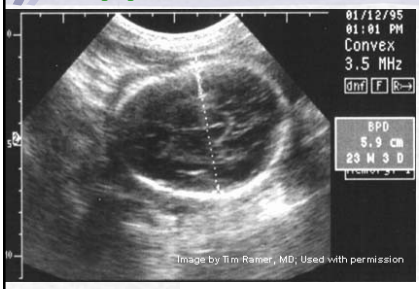


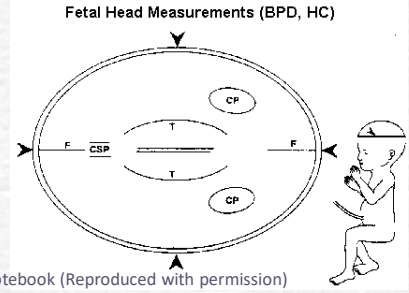
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Applications – Medical Imaging





© Family Practice Notebook (Reproduced with permission)



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Vision is confusing



"Every image is the image of a thing merely for him who knows how to read it, and who is enabled by the aid of the image to form an idea of the thing."

[Handbook of Physiological Optics by H. Helmholtz]

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Interfacing with the world



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Applications – Augmenting reality



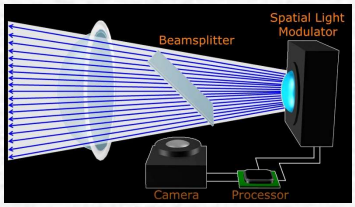
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Applications – Intelligent Devices



© Takeo Kanade, CMU



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Applications – Self drive car



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The Ultimate Goal

"...apprehension by the senses supplies after all, directly or indirectly, the material of all human knowledge, or at least the stimulus necessary to develop every inborn faculty of the mind. It supplies the basis for the whole action of man upon the outer world... For there is little hope that he who does not begin at the beginning of knowledge will ever arrive at it's end"

[The Recent Progress of the Theory of Vision by H. v. Helmholtz (1873)]

"If our long-sought quest to create autonomous anthropomorphic automata is to succeed, we must first impart perceptual abilities to machines".

[A Guided Tour of Computer Vision, by V. Nalwa]

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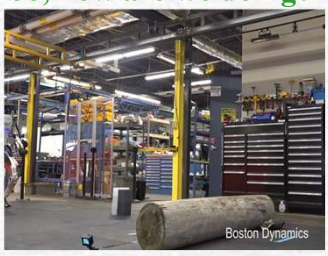
The goal of Computer Vision?

Ultimately emulating this…



So, how are we doing?







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Vision will…

- Solve harder problems with less constraints.



From http://www.telepresenceoptions.com/images/Minority_Report.gif

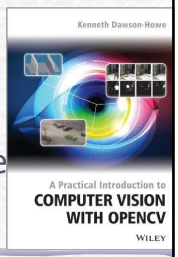
- Be a big part of User I/O
- Enhance security.
- Diagnose medical conditions reliably.
- Control vehicles automatically.
- Identify suspects through forensic analysis
- Provide more understanding about the environment.

Goals

- It is intended that the student should, at the end of the course...
  - Understand the broad subject of computer vision.
  - Understand the main algorithmic processes used to manipulate images.
  - Understand how information may be extracted from images, and the associated problems.
  - Be able to describe the various operations both algorithmically and mathematically.
  - Be able to code and test basic vision algorithms.
  - Be able to develop potential solutions to complex vision problems.

Teaching Philosophy

- Flipped classroom
  - Prerecorded lectures to be studied in advance of class.
- Teaching by example.
- Hands-on experience of computer vision operations.
- Hands-on experience of coding computer vision operations
- Application of techniques worked on in group tutorials.
- Based on a single text +
  - RANSAC + PCA + K-means + Watershed +
  - Deep Learning + …
- Lectures, slides & sample code provided online
  - On Blackboard (mymodule.tcd.ie)



Module delivery (most weeks)

- Around 4 pre-recorded lecture sessions (15-25 minutes each)
- 1 or 2 mini-tests to assess basic understanding of the pre-recorded material
  - Must be done before the Q&A session (or 0 marks).
- Q&A & 'advanced' topics session to discuss any issues/questions relating to the pre-recorded lecture material
  - Students MUST have reviewed the pre-recorded lecture sessions. If you have not reviewed them then please do not ask any questions during the Q&A session.
  - Time permitting we will look at some more advanced/applied topics in Vision such as Ethics, Text recognition, Lane Detection, Recognition using Deep Learning, Face Recognition, Head pose estimation, etc.
- Tutorial session to consider how to address real world problems using vision.
  - Students will be broken into small groups to come up with solutions (for 10-15 minutes), the course lecturer will visit the groups, and then when we come back together groups will present their solutions which will be discussed.
  - If the Q&A session does not have much to discuss, we may start the tutorial session during that hour.
- Discussion board

Module Assessment

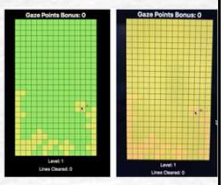
- 10% Mini-tests which assess basic understanding of the pre-recorded material.
  - Must be done before the Q&A session (or 0 marks will be awarded).
  - Single attempt only.
- 30% Programming assignment.
  - Solving a real world vision problem and assessing how you have done.
- 60% In person exam
  - 1 hour.
  - Some application of vision to real world problems type questions.
  - Some compare & contrast type questions.

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Final Year Projects

- Available now.
- Pick quickly – some areas are very popular.
- My projects:
  - Lip Reading
  - Speed camera
  - Tennis tracking
  - Cyclist evaluation
  - Pilates/yoga evaluation
  - Self drive cars on Irish roads
  - Transfer Learning
  - Anything in Computer Vision







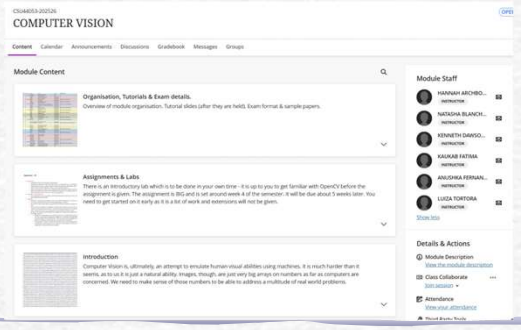


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Blackboard

- Updated to Blackboard Ultra
- Complete reorganisation
- Complete re-write of mini-tests

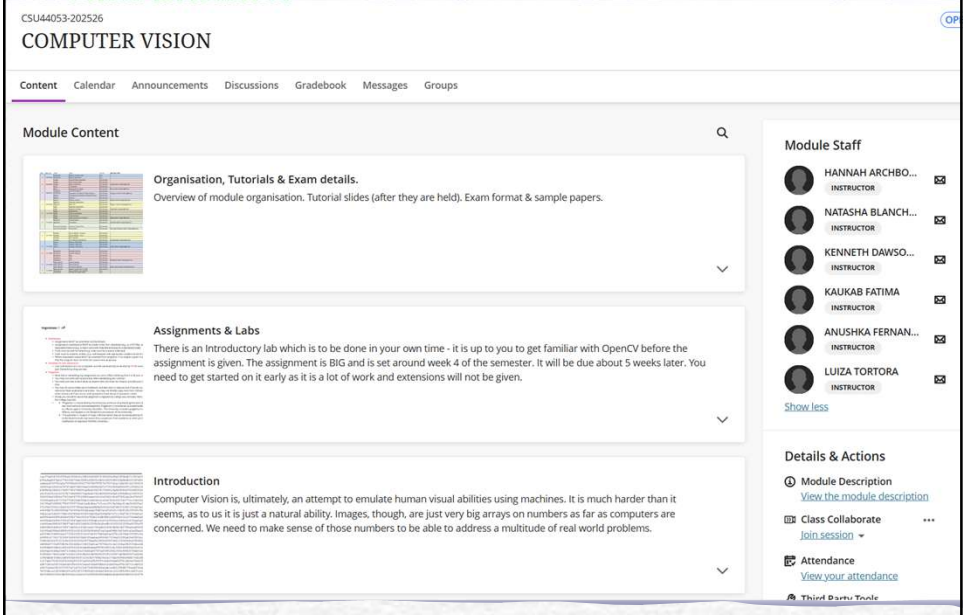
- Any issues or concerns (e.g. lecture material missing) please contact me.



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Blackboard



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Module contents

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		Conclusions	Sample exam paper tutorial	Live		Tutorial Wednesday @1pm	
							Exam (60%)

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Images

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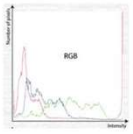
This section of the module looks at camera models (mainly the simple pinhole camera model), how images are represented (in particular sampling and quantization), the different forms of colour images (RGB, CMY, YUV, HLS), noise in images (Salt & Pepper, Gaussian), how to attenuate noise using smoothing (averaging, median filters) and finally how images can be represented at multiple scales in image pyramids.



Binary Vision

Content isn't available

Binary imaging is the processing of images with a single bit per pixel (normally represented as black and white). Topics include thresholding, threshold detection, adaptive thresholding, techniques to clean binary images (using Erosion, Dilation, Opening, Closing) and how those techniques can be applied to greyscale/colour images.



Histograms

Content isn't available

Histograms provide a way of summarizing image content in terms of the pixel values taking no account of spatial organization. Topics include 1D histograms of greyscale and colour images, 3D histograms of colour images, Histogram equalisation, Histogram comparison and (importantly) Histogram Backprojection.



Regions

Content isn't available

Region segmentation is one of the main forms of image segmentation (the other being based on edges). We look at pixel connectivity issues, the (important) Connected Component Analysis technique, then consider some of the features which can be computed for regions (e.g. Area, Minimum Bounding Rectangle, Elogatedness, Convex Hull, Moments, Concavities & Holes), and finally look at a technique for clustering colours together called k-mean clustering.

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Regions II

Content isn't available

This is a continuation of the Regions part of this module, where more advanced forms of Regions Segmentation are considered. We look at Watershed Segmentation (a reasonably intuitive technique) and then at Mean Shift Segmentation (which is much more complex but also much more powerful. Mean Shift Segmentation is particularly appropriate for segmenting colour regions.



Video

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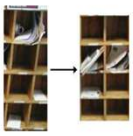
Video processing involves studying the changes in video footage - which can allow us to very simply identify where objects (such as people and vehicles) are going or what they are doing. There are a variety of background models (static, median and GMM) which are considered in this section which vary hugely in their capabilities.



Edges

Content isn't available

Edges are locations in an image where a significant change in brightness or colour occurs. There use is the other main approach to segmentation in images. First derivative edge detectors (such as Roberts, Sobel, and Prewitt) provide gradient information which must then be non-maxima suppressed and thresholded. The second derivative (such as Canny) takes a different approach to edge localisation. From edges we extract contour descriptions, or find lines or circles (using the Hough transform). To finish two different approaches to approximating lines are considered (RANSAC and least squared error).

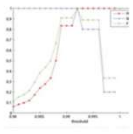


Geometric

Content isn't available

Geometric operations provide a method of transforming images to correct distortions, align images with other images or models of the world. Geometric transformations usually occur backwards (i.e. for each output point an input is found) and then a value for the output point must be interpolated from the nearby pixels in the input. Distortions which affect normal cameras are considered at the end.

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Learning & Evaluation

Content isn't available

Machine Learning is based on large amounts of sample data. It can be supervised (where the data is explicitly labelled) or unsupervised (where learning has to done based on commonality in the data). Explicit labelling is also required in evaluation where it is referred to as ground truth (and this is typically manually created). Ground truth could just be labels on images, or labelled bounding boxes on objects or even pixel based (where each pixel is labelled as a particular object or class). A variety of metrics are available to evaluate the performance of techniques (e.g. Precision, Recall, Accuracu, F1, IoU, mAP, Confusion matrices, ...).



Features

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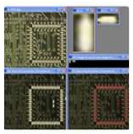
Features can be understood as being akin to corners, although they are often more complex than that. The Moravec and the Harris/Plessey corner detectors are explained. Then the Scale Invariant Feature Detector is explained as is its use for tracking and recognition.



Video II

Content isn't available

When analysing video it is often important to track objects as they move. Two techniques frequently used are mean shift tracking and optical flow.



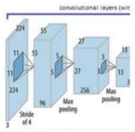
Recognition

Content isn't available

Recognition (of objects) is a key aspect of computer vision and there are numerous ways of addressing it. There are simple techniques like template matching, chamfer matching and linear classifiers (See Deep Learning section), but also more complex methods like using SIFT features (See Features section), Robust Object recognition using a cascade of classifiers and Principal Components Analysis.

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Module contents



Deep Learning in Computer Vision

Content isn't available

Deep Learning is where most of the current advances in computer vision has been happening since around 2012. It is a huge field and once which is deserved of study in its own right. This section gives an overview of the early work in this field so that students can have a framework with which to understand the current state of the art. Using traditional recognition using linear classifiers to introduce some of the issues, we move quickly to neural networks and to back-propogation (which allows them to learn weights). From there the datasets used are considered, before moving to convolutional networks (which suit image processing), residual networks, transfer learning (allowing application to different problems areas), RNNs and R-CNNs.



Extra Topics

During the module extra topics are considered which are not directly on the syllabus, in order to give a broader understanding of the area, some applications and sometimes the state of the art. Slides from these in-person sessions will be made available here.

How to approach problems in the exam

Content isn't available

Based on *A Practical Introduction to Computer Vision with OpenCV* by Kenneth Dawson-Howe © Wiley & Sons Inc. 2014